

# Zixiang Wang

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Homepage: <https://fieldry.github.io/>

Google Scholar: 333 citations

## Research Interest

I am a master's student at Peking University, advised by Prof. Liantao Ma. My research focuses on AI for Healthcare, especially medical large language models, electronic health record modeling, multi-agent collaboration, and reusable benchmark/toolkit infrastructure for clinical data analysis.

- (1) **Medical Large Language Models:** Building and evaluating LLM-driven systems for clinical prediction, reasoning, retrieval, and decision support.
- (2) **Electronic Health Record Modeling:** Modeling longitudinal, sparse, and multimodal EHR data with attention to clinical reliability and reuse.
- (3) **Benchmarks, Toolkits, and Protocols:** Creating open resources that make healthcare AI evaluation more reproducible and practically useful.

## Education

### Peking University 北京大学

2024.09 - Present

School of Software and Microelectronics · M.Eng. in Software Engineering

Working on AI for Healthcare, medical large language models, and EHR modeling under the supervision of Prof. Liantao Ma.

### Beihang University 北京航空航天大学

2019.09 - 2024.06

School of Software · B.Eng. in Software Engineering

GPA: 3.83 (Rank: 9/187, Top 5%).

Completed systematic training in software engineering and gradually shifted toward data modeling, machine learning, and healthcare AI research.

## Publications (\* indicates equal contribution; † indicates corresponding author)

### LLM Agents for Healthcare

- [WWW 2025] **ColaCare: Enhancing Electronic Health Record Modeling through Large Language Model-Driven Multi-Agent Collaboration**  
Zixiang Wang\*, Yinghao Zhu\*, Huiya Zhao\*, Xiaochen Zheng, Tianlong Wang, Wen Tang, Yasha Wang†, Chengwei Pan, Ewen Harrison, Junyi Gao†, Liantao Ma†  
*ACM International World Wide Web Conference (WWW)*, 2025
- [Preprint 2025] **HealthFlow: A Self-Evolving AI Agent with Meta Planning for Autonomous Healthcare Research**  
Yinghao Zhu\*, Yifan Qi\*, Zixiang Wang, Lei Gu, Dehao Sui, Haoran Hu, Xichen Zhang, Ziyi He, Junjun He, Liantao Ma†, Lequan Yu†  
*Preprint*, 2025
- [KDD 2024] **EHRFlow: A Large Language Model-Driven Iterative Multi-Agent Electronic Health Record Data Analysis Workflow**  
Hao Wu\*, Yinghao Zhu\*, Zixiang Wang, Xiaochen Zheng, Ling Wang, Wen Tang, Yasha Wang, Chengwei Pan, Ewen Harrison, Junyi Gao†, Liantao Ma†  
*Artificial Intelligence and Data Science for Healthcare: Bridging Data-Centric AI and People-Centric Healthcare (KDD 2024 AIDSH Workshop)*, Oral, 2024

### Toolkits & Platforms

- [CHI 2026] **Augmenting Clinical Decision-Making with an Interactive and Interpretable AI Copilot: A Real-World User Study with Clinicians in Nephrology and Obstetrics**  
Yinghao Zhu\*, Dehao Sui\*, Zixiang Wang\*, Xuning Hu, Lei Gu, Yifan Qi, Tianchen Wu, Ling Wang, Yuan Wei, Wen Tang, Zhihan Cui, Yasha Wang, Lequan Yu, Ewen Harrison, Junyi Gao, Liantao Ma†  
*ACM CHI Conference on Human Factors in Computing Systems (CHI)*, 2026
- [KDD 2026] **OneEHR: Reproducible and AI Agent-Ready Longitudinal EHR Analysis Toolkit**  
Yinghao Zhu, Zixiang Wang, Lei Gu, Dehao Sui, Yasha Wang, Ewen Harrison, Tianfan Fu, Junyi Gao, Lequan Yu†, Liantao Ma†  
*ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD-26) Hands-On Tutorials*, 2026
- [STAR Protocols 2025] **Protocol for Processing Multivariate Time-series Electronic Health Records of COVID-19 Patients**  
Zixiang Wang\*, Yinghao Zhu\*, Dehao Sui, Tianlong Wang, Yuntao Zhang, Yasha Wang, Chengwei Pan, Junyi Gao†, Liantao Ma†, Ling Wang†, Xiaoyun Zhang†  
*STAR Protocols*, 2025
- [SAIL 2025] **AIcCare: An AI-Clinician Interaction System for Transparent and Actionable Clinical Decision Support**  
Junyi Gao, Zixiang Wang, Dehao Sui, Wen Tang, Yasha Wang, Ling Wang, Liantao Ma, Yinghao Zhu  
*The Symposium on Artificial Intelligence in Learning Health Systems (SAIL)*, Best abstract nominee, 2025
- [STAR Protocols 2024] **Protocol to process follow-up electronic medical records of peritoneal dialysis patients to train AI models**  
Tianlong Wang\*, Yinghao Zhu\*, Zixiang Wang, Wen Tang†, Xinju Zhao, Tao Wang, Yasha Wang, Junyi Gao†, Liantao Ma†, Ling Wang†  
*STAR Protocols*, 2024

LLM for Healthcare

[CIKM 2024] **EMERGE: Enhancing Multimodal Electronic Health Records Predictive Modeling with Retrieval-Augmented Generation**  
Yinghao Zhu\*, Changyu Ren\*, **Zixiang Wang**, Xiaochen Zheng, Shiyun Xie, Junlan Feng, Xi Zhu, Zhoujun Li, Liantao Ma, Chengwei Pan†  
*ACM International Conference on Information and Knowledge Management (CIKM)*, 2024

[Preprint 2024] **Prompting Large Language Models for Zero-Shot Clinical Prediction with Structured Longitudinal Electronic Health Record Data**  
Yinghao Zhu\*, **Zixiang Wang**\*, Junyi Gao, Yuning Tong, Jingkun An, Weibin Liao, Ewen Harrison, Liantao Ma†, Chengwei Pan†  
*Preprint*, 2024

Healthcare Benchmark

[Preprint 2026] **Auditing medical multi-agent AI reveals risks of false consensus**  
Yinghao Zhu\*, Lei Gu\*, **Zixiang Wang**\*, Haoran Sang, Dehao Sui, Wen Tang, Lan Mi, Yasha Wang, Junyi Gao, Liang Yao, Tianfan Fu, Ewen Harrison, Lequan Yu†, Liantao Ma†  
*Preprint*, 2026

[npj Digital Medicine 2026] **ClinicRealm: Re-evaluating Large Language Models with Conventional Machine Learning for Non-Generative Clinical Prediction Tasks**  
Yinghao Zhu\*, Junyi Gao\*, **Zixiang Wang**\*, Weibin Liao\*, Xiaochen Zheng, Lifang Liang, Miguel O. Bernabeu, Yasha Wang, Lequan Yu, Chengwei Pan†, Ewen Harrison†, Liantao Ma†  
*npj Digital Medicine*, 2026

[NeurIPS 2025] **MedAgentBoard: Benchmarking Multi-Agent Collaboration with Conventional Methods for Diverse Medical Tasks**  
Yinghao Zhu\*, Ziyi He\*, Haoran Hu\*, Xiaochen Zheng\*, Xichen Zhang, **Zixiang Wang**, Junyi Gao, Liantao Ma†, Lequan Yu†  
*Conference on Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*, 2025

[Cell Patterns 2024] **A Comprehensive Benchmark For COVID-19 Predictive Modeling Using Electronic Health Records in Intensive Care**  
Junyi Gao\*, Yinghao Zhu\*, Wenqing Wang\*, **Zixiang Wang**, Guiying Dong, Wen Tang, Hao Wang, Yasha Wang, Ewen Harrison, Liantao Ma†  
*Cell Patterns*, 2024

Healthcare Modeling

[KDD 2025] **Learnable Prompt as Pseudo-Imputation: Rethinking the Necessity of Traditional EHR Data Imputation in Downstream Clinical Prediction**  
Weibin Liao, Yinghao Zhu, Zhongji Zhang, Yuhang Wang, **Zixiang Wang**, Xu Chu, Yasha Wang, Liantao Ma†  
*ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*, 2025

[CIKM 2024] **PRISM: Mitigating EHR Data Sparsity via Learning from Missing Feature Calibrated Prototype Patient Representations**  
Yinghao Zhu, **Zixiang Wang**, Long He, Shiyun Xie, Xiaochen Zheng, Liantao Ma†, Chengwei Pan†  
*ACM International Conference on Information and Knowledge Management (CIKM)*, 2024

Trustworthy AI

[KDD 2024] **RetCare: Towards Interpretable Clinical Decision Making through LLM-Driven Medical Knowledge Retrieval**  
**Zixiang Wang**\*, Yinghao Zhu\*, Junyi Gao, Xiaochen Zheng, Yuhui Zeng, Yifan He, Bowen Jiang, Wen Tang, Ewen Harrison, Chengwei Pan, Liantao Ma†, Ling Wang†  
*Artificial Intelligence and Data Science for Healthcare: Bridging Data-Centric AI and People-Centric Healthcare (KDD 2024 AIDSH Workshop)*, 2024

Selected Project

▷ **小雅医生 (AICare): AI-Powered Clinical Decision Support Platform for Electronic Health Record Analysis**, 2025  
Deployed on PKU Xplore "clinical medicine + X" Platform

Selected Honors and Awards

▷ **Outstanding Graduate of Beihang University**, Beihang University, 2024

▷ **National Encouragement Scholarship**, Beihang University, 2022, 2023

Services

▷ **Reviewer**, AAAI 2026, 2027, 2025, 2026

▷ **Reviewer**, npj Digital Medicine, 2026

▷ **Reviewer**, npj Health Systems, 2026

▷ **Reviewer**, Scientific Reports, 2026